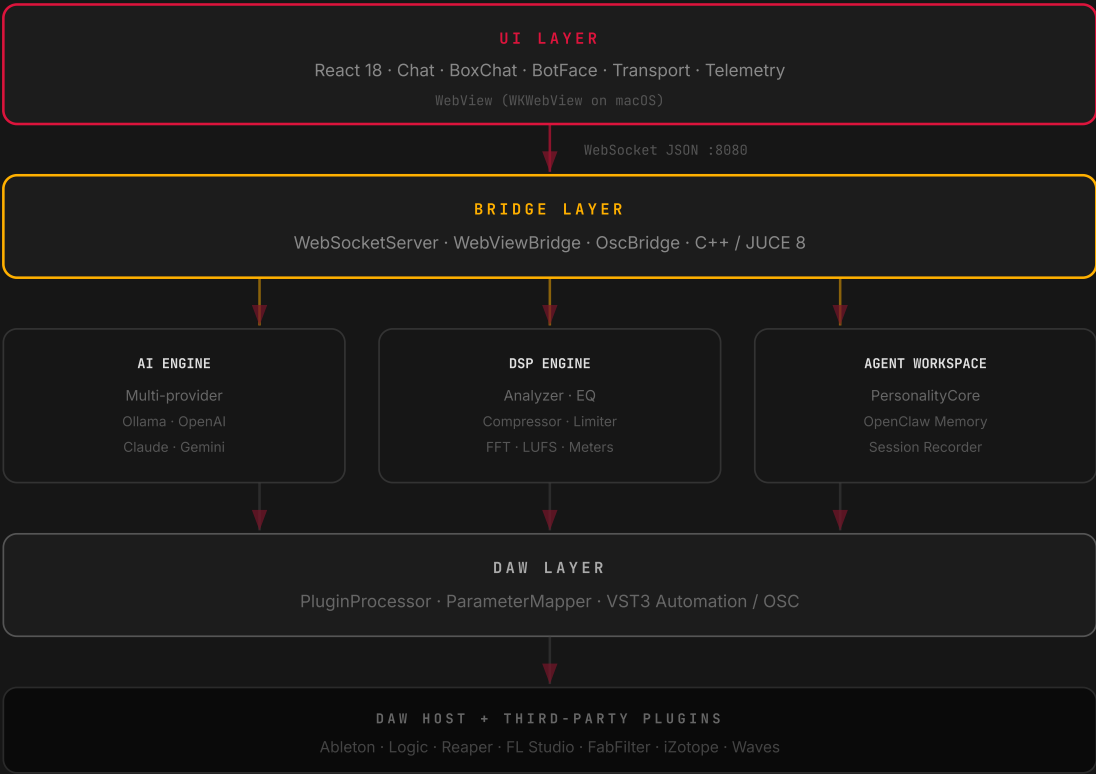


System Architecture

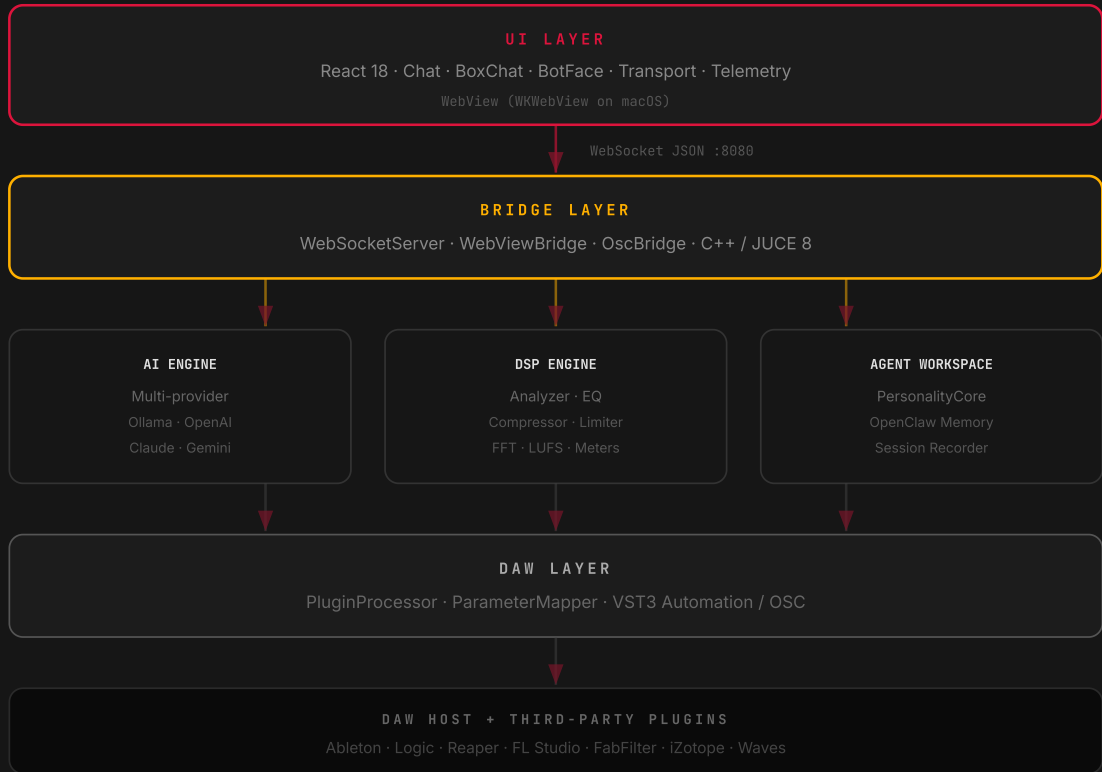
Tech Stack, Components and Data Flows

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◆ SYSTEM ARCHITECTURE



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• Tech Stack

LAYER	TECHNOLOGY	RATIONALE
UI	React 18 + Framer Motion	Reactive components, fluid animations
Styling	Tailwind CSS 4	Utility-first, zero overhead
Build	Vite 8	Fast HMR, optimised bundle
Host	JUCE 8	Industry standard for audio plugins
Bridge	WebSocket + JSON	Universal, low latency
AI	Multi-provider	Flexibility, no vendor lock-in
DSP	JUCE DSP + custom	Real-time, thread-safe
Serialisation	nlohmann/json	Header-only, fast, readable
OSC	oscpack	Standard DAW protocol

• Complete Data Flow Example

Scenario: User writes "analyze the mix and tell me what's wrong with the kick"

1. UI sends: { type:'ai.prompt', payload:{ prompt:'analyze...' } }
2. OscBridge receives, dispatches to AiEngine
3. AgentWorkspace enriches prompt:
 - BPM: 128, position: 02:34
 - Kick volume: -3dB, FFT peak at 220Hz (+2dB over ideal)
 - User memory: "prefers dry kick, narrow EQ cuts"
4. AiEngine streams response via Gemini/Claude/Ollama
5. UI renders BoxChat type 'eq' (detects "220Hz" in response)
6. User: "yes apply it"
7. ParameterMapper → FabFilter Pro-Q3: Band1 freq=220Hz gain=-3dB Q=2.1
8. AgentSoul logs: "narrow EQ cut accepted – reinforce pattern"